

Emmanouela Tsakalidou

[GitHub: EmmaTsak](#) | [LinkedIn: etsakalidou](#) | tsakalidouemmanouela@gmail.com

Software Development Graduate | Quality Engineering, Test Automation, Playwright, APIs, Databases

EDUCATION

University of Essex <i>BSc (Hons) Computing (Software Development) – Software Development, APIs, Databases, Testing</i>	September 2024 – June 2026
IVT OMIROS <i>Web Designer/Developer – Video Games</i>	September 2023 – June 2024

EXPERIENCE

Software Developer Intern <i>Valmore</i>	February 2025 – September 2025 <i>Athens</i>
- Developed, tested, and maintained application features within agile development cycles, contributing across implementation, validation, debugging, and release readiness. - Created and executed Jest test scripts to support automated test coverage and earlier identification of software defects before release. - Reproduced, investigated, and resolved software defects using structured troubleshooting and root cause analysis with developers and designers. - Built and optimized reusable React Native components while validating application behaviour, consistency, and feature integration. - Collaborated with developers and designers to investigate technical issues, communicate findings, and support reliable software delivery.	
Online Store Co-Administrator <i>Zozefina Jewelry – Little Zozefina</i>	July 2020 – October 2020 <i>Larissa</i>
- Managed and updated the online shop platform, checking customer-facing product information, prices, and images for accuracy. - Updated website content, including text, images, and multimedia elements, with close attention to consistency and user-facing quality. - Produced product photography and edited images for e-commerce listings and marketing use.	

PROJECTS

Thesis – PriceWise <i>Node.js, Express, TypeScript, React, PostgreSQL, Docker, Playwright</i>	March 2026 – Present
- Developing a full-stack grocery price comparison application covering backend services, RESTful APIs, relational database design, frontend functionality, automated data workflows, and testing. - Designed and implemented RESTful APIs with Node.js, Express, and Prisma, supporting filtering, price comparison, product aggregation, and structured data workflows. - Built React and TypeScript frontend features with responsive behaviour, multilingual support, dynamic filtering, and real-time product update flows. - Tested API responses, search functionality, filters, frontend behaviour, and real-time update flows to identify defects and verify expected application behaviour. - Used PostgreSQL and Prisma to model and validate product, supermarket, category, and pricing data through structured relational database access. - Applied Docker, Git, Playwright, PostgreSQL, and TypeScript across local development, testing, debugging, and integration workflows. - Documented technical logic, APIs, data flows, and implementation decisions to support troubleshooting, maintainability, and future development.	

SELECTED ACADEMIC COURSES

AG301 Advanced Programming	October 2025 – February 2026
- Developed and tested a RESTful API with JWT-based authentication, middleware architecture, and advanced querying. - Used Postman to validate filtering, sorting, pagination, authentication, protected endpoints, and API responses.	

- Structured backend routes to support issue isolation, maintainability, API security foundations, and testable application behaviour.

Full-Stack, Database, and Software Engineering Foundations

September 2024 – June 2026

- Applied Python, PHP, JavaScript, TypeScript, Java, C#, C, C++, SQL, database querying, software design, testing, and debugging principles across academic programming tasks.
- Worked with relational and non-relational database concepts, API design, authentication, frontend development, software testing, and structured problem solving.

SKILLS

Testing and Quality: Jest, Playwright, Postman, API Testing, Test Scripts, Debugging, Troubleshooting, Root Cause Analysis, Regression Testing Foundations

Full-Stack Development: React, React Native, Node.js, Express, RESTful APIs, HTML/CSS, Responsive Web Interfaces

Programming: Python, PHP, TypeScript, JavaScript, Java, C#, C, C++, SQL

Databases: MySQL, PostgreSQL, MongoDB, Prisma, Relational Databases, NoSQL Databases, Database Querying, Data Filtering, Data Validation

Tools and Infrastructure: Git, GitHub, Docker, Playwright, Socket.IO, Unix/Linux Command Line Foundations, WordPress, Figma

Software Engineering: Agile Development, Software Development Lifecycle, Technical Documentation, Defect Investigation

Languages: English (Fluent), Greek (Native)

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Certifications: Learn SQL Course – Codecademy (2024); React Native – The Practical Guide – Udemy (2025); Responsive Web Design – freeCodeCamp (2023)

Professional Development: Global Game Jam Larissa (2024); Aegean College Leadership Conference (2024); Aegean College Success Stories Conference (2023); 11th Student Festival of Digital Creation (2019)